

WATERWALKER · DESIGN DECISION RECORD

WaterWalker TRL 2 review decisions

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 6 and the cross-cutting items; items 9 to 13 accepted later on 2026-09-25 in WWK-DDR-002); items 7 and 8 remain proposed, awaiting Amish

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-24) listed seven items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." He also approved three cross-cutting additions to the SwapCell interface (issued as SwapCell interface v0.3) and a portfolio pricing rule for shared SwapCell packs.

This record lists what that instruction decides and what it leaves open. Items with no recommendation stay open, and so do the new engineering proposals that came out of the TRL 3 calculations (WWK-CAL-001). TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for items 1 to 7 are in docs/REVIEW.md (2026-09-24) and in the key design choices of WWK-PRC-001 v0.2. They are not repeated here.

■ Decision

Each row in Table 1 is "Decided by Amish, 2026-09-25: go with recommendation".

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Wheel size	Decided by Amish, 2026-09-25: go with recommendation. 26 x 2.1 to 2.4 in wheels all round	WWK-PRC-001 v0.3, WWK-REQ-001 R12, <code>cad/src/model.py</code>
2	Assist in the first prototype	Decided by Amish, 2026-09-25: go with recommendation. The first prototype has no assist but keeps mounting points for it; the \$450 budget is kept. The budget now covers the unassisted first prototype with mounting points; the assist kit is a later second prototype	WWK-REQ-001 R11 and R13, <code>bom/bom.csv</code> , <code>project.yaml</code> (budget unchanged)
3	Steering	Decided by Amish, 2026-09-25: go with recommendation. Full-swivel lockable front casters	WWK-PRC-001 v0.3, WWK-DWG-001
4	User position	Decided by Amish, 2026-09-25: go with recommendation. Between the rear wheels	WWK-PRC-001 v0.3, WWK-DWG-001
5	Cradle and clay pots	Decided by Amish, 2026-09-25: go with recommendation. Four 20 L jerrycans or two 20 L clay pots per trip, until co-design shows how often pots are used. The review said the pitch should then be reworded; the pitch and README now read "four 20 L jerrycans or two clay pots"	WWK-REQ-001 R1, <code>project.yaml</code> , <code>README.md</code>
6	Brake type	Decided by Amish, 2026-09-25: go with recommendation. Drum brakes on both rear wheels with a parking latch. Whether users want a dead-man brake is a question for co-design (item 8)	WWK-PRC-001 v0.3, WWK-REQ-001 R10

Cross-cutting approvals from the same instruction, recorded here as they apply to WaterWalker:

- **SwapCell interface v0.3.** Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell interface adds a wake method for hosts without CAN (item W), a charge-while-discharging mode (item C) and a latch vibration rating for vehicles (item V). WaterWalker's assist mounting points are drawn to SwapCell interface v0.3: a class V1 vehicle receiver on a mounting plate above the tires, with the pack's back and lid faces open to air, and a 10 kΩ INTERLOCK coding resistor in the receiver so a controller without CAN can wake the pack (item W) and draw legacy discharge-only current (15 A limit). WaterWalker does not use item C: the carrier never charges the pack. New requirement R13 (WWK-REQ-001).
- **Shared packs are priced once.** Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell pack is priced in the SwapCell BOM (about \$414) and excluded from this repo's budget. BOM item 10 carries a zero price with that note.
- **Co-design partners.** Decided by Amish, 2026-09-25 (cross-cutting): community designs pick co-design partners per area later. The partner stays open (item 7).

Items left open by this record

Items 7 and 8 had no recommendation in the TRL 2 review and remain **Proposed, awaiting Amish**. Items 9 to 13 were new proposals from WWK-CAL-001; on 2026-09-25 Amish accepted all recommendations ("i accept all your recommendations, go with them across all repos"), so each is now **Decided by Amish, 2026-09-25: go with recommendation**, as recorded in WWK-DDR-002.

#	Item	Options	Recommendation and status
7	First co-design partner and region	Helpful Engineering network, an NGO or a university in a region with sandy routes	None; for Amish to decide, per the portfolio rule on partners. Proposed, awaiting Amish
8	Dead-man brake	Ask users in co-design; fit a spring-applied brake released by the hip bar; or rely on the lever and latch	None until users are asked. Proposed, awaiting Amish
9	Rear wheel mounting	Keep the TRL 2 cantilever stub axles (a standard M10 axle reaches about 472 MPa at 2.5 g, above a bicycle axle's strength); or hold each rear wheel in a fixed rigid bicycle fork with 100 mm dropouts (about 89 MPa)	Fixed forks , as modelled. This needs 100 mm front-type drum hubs on the rear wheels and a track of 770 mm (was 800 mm) to stay within the 900 mm width. Decided by Amish, 2026-09-25: go with recommendation (WWK-DDR-002)
10	Caster arm and front riser position	Keep the TRL 2 riser position (the swivelling tire hits the riser); or move the risers back to x = 1,150 mm, lengthen the wheelbase to 1,530 mm and shorten the cradle to 780 mm	Move the risers , as modelled. Two clay pots then fit with 8 mm to spare (12 mm with the thinner cradle walls). Decided by Amish, 2026-09-25: go with recommendation (WWK-DDR-002)
11	Main frame section	30 x 30 x 1.5 mm square (yield factor 1.4 at 2.5 g); 40 x 30 x 1.5 mm rectangular (2.1); 40 x 30 x 1.2 mm (1.7, 1.8 kg lighter)	40 x 30 x 1.5 mm for the prototype, as modelled. Decided by Amish, 2026-09-25: go with recommendation (WWK-DDR-002); the 40 x 30 mm section is kept and its wall is 1.2 mm under item 12
12	Mass (R9 not met at 40.8 kg)	Apply the four mass options in WWK-CAL-001 (to about 35.6 kg, still over 35 kg); relax R9; or both	Apply the thinner cradle and the 1.2 mm wall, keep the puncture protection, and revisit R9 with users in co-design. Decided by Amish, 2026-09-25: go with recommendation (WWK-DDR-002); applied to the model, empty mass now 37.4 kg
13	Parking brake	Lever latch alone (about 189 N hand force at the latch); a positive lock pin through a rear wheel or drum; wheel chocks on loose ground	Add a positive lock pin and keep the lever latch for short stops. Decided by Amish, 2026-09-25: go with recommendation (WWK-DDR-002); BOM item 15

■ Consequences

- The first prototype is priced at about \$430 against the \$450 budget (R11 at risk, \$20 margin). The optional assist kit (motor, sensor, receiver) adds about \$260, outside the first-prototype budget, plus a shared SwapCell pack.
- The requirement set changes as follows (WWK-REQ-001 v0.3): R1 is redefined to four jerrycans or two clay pots; R11 is redefined to the unassisted first prototype with mounting points, excluding the SwapCell pack and the assist kit; R13 is added for assist readiness to SwapCell interface v0.3.
- The pitch in `project.yaml` and the README now says "four 20 L jerrycans or two clay pots" and calls the assist a later option with mounting points in the first prototype.
- TRL 4 (building and testing the prototype) is on hold by Amish's instruction.
- Later on 2026-09-25 Amish accepted the recommendations for items 9 to 13 (WWK-DDR-002). The first prototype is now about \$426 and 37.4 kg empty (WWK-CAL-001 v0.2).